

Aliko ConTech — Developer Workflow Documentation

Subdomain: contech.alikohub.com

Purpose: Internal digital construction management platform for **Aliko Construction projects only**, with controlled client visibility.

Status: Simplified MVP aligned with AlikoHub v1.0 architecture (SSO + RBAC already in place).

1. Purpose & Scope

Aliko ConTech is a **private construction management platform** built exclusively for **Aliko Construction**, designed to digitize internal workflows and provide transparent project visibility to clients.

This version of Aliko ConTech is **not a public SaaS, does not support external contractors**, and **does not include payment processing**.

The platform's primary objectives are:

- Digitize Aliko Construction's internal project workflows
- Enable real-time communication between Aliko contractors and clients
- Provide clients with structured, read-only oversight of their construction projects
- Maintain simplicity for non-technical users while remaining scalable for future expansion

2. User Roles & Access Model (RBAC)

Only **three roles** exist in this version.

2.1 Admin (Aliko ConTech Admin)

Route: [/admin](#)

Admin represents Aliko Construction management.

Permissions:

- Create and manage **clients**
- Create and manage **contractors**
- Create and manage **projects**
- Assign contractors and clients to projects
- View all projects and dashboards
- Monitor overall progress across all projects
- Access all uploaded documents, photos, and reports
- Manage system-level settings (future)

2.2 Contractor (Internal Aliko Contractor)

Route: </contractor>

A contractor is an **Aliko Construction staff member** responsible for overseeing one or more projects.

Permissions:

- View only projects they are assigned to
- Update project progress
- Manage tasks and milestones
- Upload site photos and documents
- Communicate with clients via project communication tools
- View client messages and respond
- Cannot create or delete projects or users

Each contractor:

- Has a unique login created by Admin
- Can be assigned to multiple projects
- Cannot access projects they are not assigned to

2.3 Client (External Client)

Route: </client>

A client is an external customer who hired **Aliko Construction**.

Permissions:

- View only their own projects
- Track progress, milestones, and timelines

- View uploaded photos, documents, and reports
- Communicate with the assigned contractor
- Cannot edit project data
- Cannot see internal contractor-only notes

Clients:

- Are created by Admin after engagement begins
- May have **multiple projects**
- Only see what Admin and Contractor expose to them

3. High-Level Platform Workflow

3.1 Client Onboarding Flow

1. A potential client contacts Aliko Construction (offline or via website)
2. Admin creates a **Client account** in ConTech
3. Login credentials are sent to the client
4. Client gains access to [/client](#) dashboard
5. Client sees:
 - Welcome screen
 - Empty project list (if no project yet)
 - Contact details and guidance

No self-signup for clients in this version.

3.2 Contractor Onboarding Flow

1. Admin creates a **Contractor account**
2. Contractor receives login credentials
3. Contractor logs in via SSO
4. Contractor sees [/contractor](#) dashboard
5. Dashboard initially shows:
 - Assigned projects
 - Notifications
 - Pending tasks (if any)

3.3 Project Creation Flow (Admin)

1. Admin logs into [/admin](#)
2. Creates a new project:
 - Project name (e.g., “G+5 Residential Building – Ayat”)
 - Project description
 - Client assignment
 - Contractor assignment
 - Start date / estimated end date
3. Project becomes visible:
 - To assigned contractor
 - To assigned client
4. Project status initialized as “**Active**”

4. Dashboards & Core Features

4.1 Contractor Dashboard ([/contractor](#))

Main purpose: **Project execution + communication**

Key sections:

a) Project List

- All assigned projects
- Status indicators (Active / On Hold / Completed)
- Quick links to each project

b) Project Detail View

For each project:

- Overview (name, client, timeline)
- Progress indicator (% or milestone-based)
- Task & milestone list
- Uploaded documents & photos
- Client communication panel

c) Progress Updates

Contractor can:

- Update progress percentage or milestones
- Add textual updates (weekly summaries)
- Upload site photos (dated & tagged)

- Upload documents (reports, drawings, approvals)

d) Client Communication

- Threaded messages per project
- Contractor ↔ Client messaging
- Attachments supported
- Messages are logged and timestamped

4.2 Client Dashboard (/client)

Main purpose: **Transparency & trust**

Key sections:

a) Project List

- All projects belonging to the client
- Status & progress indicators

b) Project Detail View

Client can see:

- Project overview
- Progress timeline
- Uploaded photos
- Shared documents
- Contractor updates
- Message contractor

Client **cannot**:

- Edit project data
- Upload files (unless allowed later)
- See internal notes

4.3 Admin Dashboard (/admin)

Main purpose: **Full oversight & control**

Key sections:

a) System Overview

- Total projects
- Active projects
- Contractors count
- Clients count

b) Project Management

- Create/edit/archive projects
- Assign contractors
- Assign clients
- View project-level activity logs

c) User Management

- Create contractors
- Create clients
- Reset credentials
- Deactivate users

d) Monitoring & Reporting

- View progress across all projects
- Identify stalled or delayed projects
- Access all communications and uploads

5. Communication & Data Flow

5.1 Communication Rules

- All communication is **project-scoped**
- Messages are persistent (audit trail)
- Admin can view all communications
- No external chat integrations (simple internal messaging)

5.2 File & Media Handling

- Photos and documents uploaded by contractors
- Files stored securely (S3 / Cloudinary)
- Clients have read-only access

- Files categorized (Photos, Reports, Drawings, etc.)

6. Contact Form (Public)

6.1 Purpose

Allow potential clients to contact Aliko ConTech without logging in.

6.2 Behavior

- Simple form:
 - Name
 - Email
 - Phone
 - Message
- On submit:
 - Email sent to ConTech admin inbox
 - No database persistence required for MVP

7. Non-Goals (Explicitly Excluded)

The following are **not included** in this version:

- Payment processing
- External contractors
- Inspector or project manager roles
- Public project listings
- Client self-registration
- Advanced analytics
- Mobile app (web responsive only)

8. Simplicity & UX Principles

- Minimal screens
- Clear labels (non-technical language)
- Step-by-step flows
- Large buttons and readable text

- Mobile-friendly layouts
- Avoid nested complexity

This platform must be usable by:

- Construction staff
- Non-technical clients
- Admins managing multiple projects

9. Future Scalability (Not Implemented Now)

The architecture **must allow** future expansion:

- External contractor onboarding
- Payment integration
- Inspector roles
- Multi-company usage
- Advanced reporting

But **no UI or logic should expose these yet.**

10. Summary

New Aliko ConTech is:

- Internal-first
- Simple
- Transparent
- Secure
- Scalable by design

It exists to **digitize Aliko Construction operations today**, while laying the groundwork for a broader construction tech platform tomorrow.